Lesson Planning

Complete Lesson Plan Book

Class: Class 10

Subject: Biology

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Cell: The Basic Unit of Life

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Unit 1: Cell: The Basic Unit of Life

Learning Objectives

• Students will state the cell theory and explain what it means.

• Students will identify the main parts of a cell and describe the function of each.

• Students will compare a plant cell with an animal cell.

Key Concepts

• A cell is the smallest unit of life; all living things are made of cells.

• The cell theory states that all organisms are made of cells and that new cells come from existing cells.

• The nucleus controls the cell's activities and carries the genetic material.

• The cytoplasm is the jelly-like substance where most chemical reactions of the cell take place.

• The cell membrane controls what enters and leaves the cell; plant cells also have a cell wall and chloroplasts.

Teaching Methodology (Step by Step)

1. The teacher asks students what the smallest building block of a house is, then compares it with the cell in a living body.

2. The teacher shows a diagram of a cell and explains that every living thing is made of cells.

3. The teacher states the three points of the cell theory and gives a simple example for each.

4. The teacher explains the nucleus, cytoplasm and cell membrane one by one with their functions.

5. The teacher draws a plant cell beside an animal cell and points out the cell wall, chloroplast and large vacuole.

6. The teacher asks students why a single cell can still be called alive, and discusses their answers.

Classroom Activities

• Label the cell: students label a diagram of an animal cell with nucleus, cytoplasm and cell membrane.

• Plant or animal: students sort cell features into plant-only, animal-only, or both.

• Microscope view: students look at a prepared slide or a printed micrograph and sketch one cell they see.

Assessment Questions

1. State the cell theory in your own words.

2. Name the three main parts of an animal cell and write one function of each.

3. Give two differences between a plant cell and an animal cell.

Homework

• Draw a labelled diagram of a plant cell and an animal cell and list three differences.

• Write five sentences explaining why the cell is called the basic unit of life.

Time Allocation

40 minutes